Quang Van Tran

PhD Researcher in Civil Engineering · The University of Texas at San Antonio
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RESEARCH INTERESTS

Urban climate and heat inequality; historical redlining; satellite remote sensing and geospatial analysis; computational fluid dynamics; deep reinforcement learning; indoor environmental modeling; and building energy optimization.

EDUCATION

PhD in Civil Engineering · University of Texas at San Antonio, USA · 2025-present
Supervisor: Prof. Farzad Hashemi

MEng in Architectural Engineering · Kyung Hee University, Republic of Korea · 2021-2023
Supervisor: Prof. Geun Young Yun

BE in Environmental Engineering · Ho Chi Minh City University of Natural Resources and Environment, Vietnam · 2013-2017

RESEARCH EXPERIENCE

Research Assistant, University of Texas at San Antonio · 2025-present
Urban heat and social vulnerability; environmental monitoring and remote sensing; neighborhood and residential airflow and heat modeling.

Research Assistant, Kyung Hee University · 2021-2023
CFD and machine learning for indoor airflow, daylighting analysis, and reinforcement learning for building energy performance.

Student Assistant, HCMUNRE-Kyushu University · 2016-2019
Numerical airflow simulation and dynamic meshes for biological airway research.

SELECTED JOURNAL ARTICLES

Year-Round Heat Exposure and Structural Spatial Vulnerability: A Case Study of San Antonio, Texas (2025). *Sustainable Cities and Society*. DOI

Online Transfer Learning for Accelerating Deep Reinforcement Learning for Building Energy Management (2024). *Journal of Building Performance Simulation*. DOI

AI Management Platform for Privacy-Preserving Indoor Air Quality Control: Review and Future Direction (2024). *Journal of Building Engineering*. DOI

Predicting Indoor Temperature Distribution in Naturally Ventilated Office Room Using Long Short-Term Memory Networks (2023). *Architectural Engineering and Design Management*. DOI

Recent Advances and Effectiveness of Machine Learning Models for Fluid Dynamics in the Built Environment (2024). *International Journal of Modelling and Simulation*. DOI

Data-Driven Prediction of Indoor Airflow Distribution in Naturally Ventilated Residential Buildings Using Combined CFD Simulation and Machine Learning (ML) Approach (2024). *Journal of Building Physics*. DOI

Using Deep Learning to Optimize the HVAC System in the Residential Building (2024). *Journal of Green Building*. DOI

MANUSCRIPTS UNDER SUBMISSION

Why Smart Cities Fail Equity and How AI-BIM Integration Can Fix It: From Review to Framework (2026). *Submitted manuscript (not peer reviewed).*

UrbanDRL-CFD: A Prospective Framework for Integrating Computational Fluid Dynamics and Deep Reinforcement Learning (2026). *Submitted manuscript (not peer reviewed).*

SELECTED CONFERENCE PAPERS AND CONTRIBUTIONS

Surface-Heat Disparities across Historically Redlined U.S. Cities Vary by Climate, Income, and Time of Day (2026). AGU Annual Meeting 2026, San Francisco, CA. *[Submitted]*

Roof Retrofit Scenarios for Thermally Vulnerable Homes: A CFD-Based Evaluation of Indoor Comfort and Air Quality in San Antonio, Texas (2026). 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings (IAQVEC), Los Angeles, CA. DOI

Data-Driven Urban Heat Mitigation: Integrating CFD and Machine Learning for Adaptive Cooling Strategies (2025). 12th International Conference on Urban Climate, Rotterdam, Netherlands. DOI

Efficient Prediction of Indoor Airflow in Naturally Ventilated Residential Buildings Using a CFD-DNN Model Approach (2023). 2nd International Conference on Green Building, Civil Engineering and Smart City, China. DOI

SELECTED HONORS AND AWARDS

- IAQVEC 2026 Scholarship Award, University of Southern California (2026)
- UTSA Graduate School Academic Travel Fund (2026)
- UTSA School of Architecture and Planning Research Travel Award (2026)
- Fully funded PhD scholarship, UTSA (2025-2029, as listed in source CV)
- Presidential Scholarship, Kyung Hee University (2021-2023)
- SAKURA Science Program, Kyushu University (2017)

ACADEMIC SERVICE AND TECHNICAL SKILLS

Service: Program coordinator and session moderator, UTSA School of Architecture and Planning research colloquium (Fall 2026). Peer reviewer for scholarly journals in the built environment and environmental modeling. AAG and AGU member.

Tools: Python, OpenFOAM, ANSYS Fluent, Rhinoceros / Grasshopper, ArcGIS Pro, Google Earth Engine, SPSS, Linux, OpenLCA.